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Ciclo: 26-1
Materia: Terminología Especializada en Tecnología Inglés-Español
Curso: IT0627
Licenciatura: Interpretación y Traducción
Grupo: 6A

HW_03 Science Basic Vocab (convert to 10 point scale)

Criteria	Excellent	Satisfactory	Needs Improvement
Task 1: Definitions System = 8 Int. Prop. = 4 Ext. Prop. = 6 Con. Laws = 5 Total = 23	Complete definitions: System (open, closed, isolated, boundary, surroundings, interactions/flows, state, properties) Intensive property (temperature, pressure, velocity, density) Extensive property (mass, charge, linear momentum, angular momentum, energy, and entropy) Conservation laws (mass, charge, linear momentum, angular momentum, energy)	Partial definitions	Incomplete and missing definitions
Document Formatting and Presentation	Document is perfectly formatted according to Marron Style Guide .	Minor formatting errors present but overall follows Marron Style Guide.	Formatting does not follow the Marron Style Guide or is inconsistent throughout.

Assignment:

***HW_03 [Due: 07 Oct 2025]

-- Task 1

Write extended and complete definitions (in both English and Spanish) for the following terms and concepts:

System

(open, closed, isolated, boundary, surroundings, interactions/flows, state, properties)

Intensive property

(temperature, pressure, velocity, density)

Extensive property

(mass, charge, linear momentum, angular momentum, energy, and entropy)

Conservation laws

(mass, charge, linear momentum, angular momentum, energy)

-- Task 2

Work through the Study Guide for Exam 1P (below). You do not need to answer the study guide questions in your homework, just be sure you can answer the questions on the exam.

Study Guide for Exam 1P

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1. Be able to write a condensed summary from your paper on the history of a scientific discipline in Mexico. Your condensed summary should certainly recount the history of the scientific discipline you selected, but it must focus on the present state of the discipline in Mexico. You should be able to write this condensed summary in both English and Spanish.
 2. Describe the mental process that engineers and scientists use to solve a problem. More specifically,
 - a. Given a design or conceptual problem, what is a first step that engineers and scientists are likely to take? Why?
 - b. What are the accepted foundations for conceptualizing a problem? That is, what is the worldview or the overarching mental model that lies at the very foundation of science and engineering? What are the components of this worldview?
 - c. What ASSUMPTIONS are made, consciously or not, when engineers and scientists begin modeling a problem?
 - d. When are models useful for reaching a solution to a problem? If no solution presents itself what can the engineer or scientist do?
 3. Jargon is defined as the set of special words or expressions that are used by a particular profession or group and are difficult for others to understand. Answer the following related to jargon:
 - a. Why does jargon evolve in virtually any human endeavor?
 - b. Is scientific language considered jargon? Explain.
 - c. Compare and contrast “good” and “bad” jargon.
 4. The following is a description of a simple system

A metal pot filled with water is placed on a stove and heated. The system is defined as the metal pot plus the water. As the water heats up, some of it evaporates into the air. The temperature of the water increases, and the volume of the remaining liquid decreases.

- a. Is this system open, closed, or isolated? Explain your reasoning.
- b. What is entering the system? What is leaving the system? What remains unchanged in the system?
- c. Identify whether each of the following properties is intensive or extensive. Justify your answer.
 - Temperature
 - Mass of the water
 - Density of the water
 - Volume of the water

5. Give a definition of the following terms in both English and Spanish:

- mass
- charge
- linear momentum
- angular momentum
- energy

6. Be prepared to translate the abstract from a scientific paper from English to Spanish and/or vice versa. For example, select either the English or Spanish version of the abstract below. Translate to the second language and compare your translation to the one provided by the authors..